

Ava Frohna

+61 466 644 854 • frohna.ava@gmail.com • Sydney, Australia • [LinkedIn](#) • [GitHub](#)

PROFILE

Cybersecurity Technical Lead and co-founder of NORA, an AI healthcare startup that reached the global top 20 of the Harvard Health Systems Innovation Lab venture program, and Master of Computer Science (Cybersecurity) student at the University of Sydney. Combines full-stack engineering and machine learning with governance, risk, and compliance depth, ISO 27001, ISO 14971, IEC 62304, and TGA groundwork, translating technical vulnerabilities into actionable risk decisions for products where patient safety and commercial viability drive the design.

EXPERIENCE

Technical Lead & Co-Founder – NORA, Sydney *April 2026 – Present*

- Own the full technical surface of a clinical communication platform that helps nurses walk into the room prepared and breaks down patient language barriers: Fastify/PostgreSQL backend, React nurse dashboard, React Native (Expo) patient app, and provider-agnostic AI adapters, deployed on Microsoft Azure
- Serve as the sole technical lead in a three-person team of clinicians, engineers, and operators, translating clinical, business, and regulatory requirements into technical product decisions and communicating trade-offs that shape strategy, usability, and patient safety
- Built the compliance scaffolding for a regulated medical-software pathway, ISO 14971 risk management, ISO 27001 ISMS, and IEC 62304 software lifecycle, laying the groundwork for TGA submission
- Communicating technical trade-offs to non-technical teammates to support product strategy, usability decisions, patient safety, cybersecurity, and future commercial deployment
- **Took NORA from founding to the global top 20 (of 1,500 teams) at the Harvard Health Systems Innovation Lab Demo Day in ten weeks**, 1st place at the Sydney hub, then representing Australia through the venture building program

Engineering Research Intern (Security Research) – University of Sydney *Jun 2026 – Present*

- Conducting security research with a supervising professor in the Faculty of Engineering, exploring risk in agentic AI systems
- Reviewing the emerging threat and risk landscape around agentic AI to help scope the research direction, drawing on prior work in vulnerability analysis and risk assessment

Web Developer – You Belong San Diego (non-profit) *Jun 2026 – Present*

- Designing and building the website for a non-profit that helps people in San Diego find groups and communities of like-minded people
- Helping establish the organisation's social media accounts and online presence to support community outreach

Mobile Expert – T-Mobile, San Diego *May 2025 – Jan 2026*

- Diagnosed and resolved complex device issues, translating technical solutions for non-technical customers to drive first-contact resolution during high-volume periods.
- Trained and supported new team members on customer-service workflows, device troubleshooting, and internal tools.

PROJECTS

AI-Based Clinical Decision Support System – Pilot Deployment Plan *Feb – May 2026*

INFO6007 Project Management in IT, University of Sydney – Best Project Award

- Led a five-person team as Project Manager, producing the complete development and pilot-deployment plan for an AI-based clinical decision support system (CDSS) integrating with hospital EMR systems

- Owned the project charter, governance structure and RACI, scope and WBS, critical-path schedule, and a costed AUD \$650K budget with phased funding releases and contingency reserves
- Developed the risk register, quality plan (PDCA/DMAIC), supplier evaluation, communication and change-management plans, and earned-value (EVM) monitoring, with cybersecurity, privacy compliance, and clinician adoption treated as first-class workstreams

Open-Source Exploit Prediction Scoring System (EPSS) – [GitHub](#)

Feb – Jul 2025

Bachelor Thesis, Vrije Universiteit Amsterdam

- Designed a data-ingestion pipeline using AI scrapers to collect and process ~9,600 CVEs, combining CVSS metrics, community threat intelligence, and social-media signals with SentenceBERT embeddings into a comprehensive feature set.
- Trained and evaluated a neural-network classifier achieving over 90% accuracy in predicting vulnerability exploitation likelihood, validating open-source threat intelligence for automated risk assessment.
- Published as an open-source alternative to proprietary vulnerability scoring, collaborating in a four-person team across data collection, feature engineering, modelling, and research documentation.

EDUCATION

Master of Computer Science (Major: Cybersecurity) – University of Sydney *Feb 2026 – Nov 2027*

Relevant Coursework: Introduction to Cybersecurity, Project Management, Professional Practice in IT, Cybersecurity

Clubs: Women in Technology, Cyber Security Society, Women in Engineering Society

Bachelor of Computer Science – Vrije Universiteit Amsterdam

Aug 2022 – Jul 2025

Relevant Coursework: Operating Systems, Computer Organization, Data Structures and Algorithms, Security and Safety Engineering, Automata and Complexity, Digital Health Software, Cloud Computing

Exchange semester: The University of Queensland, Brisbane (Information Security, Cloud Computing, Mobile Programming)

SKILLS

Security & GRC: Cyber risk management, threat modeling, vulnerability assessment, cryptography, secure software development, ISO 27001, ISO 14971, IEC 62304, TGA regulatory groundwork

Languages: Python, TypeScript, JavaScript, C, C++, Java, SQL

Web & Mobile: React, React Native (Expo), Fastify (Node.js), PostgreSQL, secure API design

Cloud & Systems: Microsoft Azure, Google Cloud Platform, Docker, Kubernetes, Unix/Linux, distributed systems

AI & ML: Neural networks, embeddings, LLM integration, prompt engineering, AI-assisted development

Project & Professional: Project management (WBS/CPM, EVM, risk registers), stakeholder communication, cross-functional leadership, mentoring, leadership, teamwork, problem solving

AWARDS

Best Project Award – INFO6007 Project Management in IT, University of Sydney (Jun 2026)

Certificate of Achievement, Global Top 20 Demo Day Finalist – Harvard Health Systems

Innovation Lab Venture Building Program (Jun 2026)

1st Place – Harvard Health Systems Innovation Lab Hackathon, Sydney Hub (Apr 2026)